BIS629: Homework 1

Due on the start of class on Sept 29, 2026

Note: Please include well-labeled computer code and output at the end of this homework, but do not include computer code and output as answers to homework questions. PASS is not needed for this homework, please use R for your calculation.

The use of AI tools is encouraged. Students must submit the following when using AI:

a. State how AI has been used, the name of the software and version;

b. In an Appendix, provide your prompts;

c. In the body of the homework, show how you verified the accuracy of any AI-generated information, reasoning, or output used in your homework.

Students are responsible for the accuracy of all submitted work. Please note that AI makes many mistakes, and AI-generated content should not be accepted without independent verification.

1a. Prove that the "classic" test for the difference between two sample means, $H_0 : \mu_1 = \mu_2$, is the same as the likelihood-based Wald test for this same hypothesis, when $Y_j \sim N(\beta_0 + \beta^* X_j, \sigma_1^2 + X_j(\sigma_2^2 - \sigma_1^2))$ for $j = 1, ..., n_1 + n_2$, and the sample variances are known. Be sure to use the likelihood given here, parameterized by $(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2)$ as the basis for this derivation. Here, X is a binary variable with value 1 if participant j is in the intervention group(group 2), and 0 otherwise, and $\sum_{j=1}^{n_1+n_2} X_j = n_2$. Begin by specifying H_0 in terms of β_0 and β^*.

1b. Now, show under what conditions, if any, the likelihood-based Wald test for these data

continues to have the same form as the "classic test" when the sample variances are not known.

Note: When the sample variances are known, the "classic test" has form

$$Z^2 = \frac{(\hat{\mu}_1 - \hat{\mu}_2)^2}{\sigma_1^2/n_1 + \sigma_2^2/n_2},$$

and when the sample variances are not known, the "classic test" has form

$$Z^2 = \frac{(\hat{\mu}_1 - \hat{\mu}_2)^2}{\hat{\sigma}_1^2/n_1 + \hat{\sigma}_2^2/n_2},$$

where $\hat{\sigma}_1^2$ is the standard sample variance estimator, $\hat{\sigma}_1^2 = \frac{\sum_{j=1}^{n_1+n_2}(1-X_j)(Y_j-\hat{\mu}_1)^2}{n_1-1}$, $\hat{\mu}_1 = \frac{\sum_{j=1}^{n_1+n_2}(1-X_j)Y_j}{n_1}$, and likewise for $\hat{\sigma}_2^2$.

2. Show that the "classic" χ^2 test for the difference between two proportions, $H_0 : p_1 = p_2$, is the likelihood-based Wald test when $Y_j \sim Bern(\beta_0 + \beta^* X_j)$ for $j = 1, ..., n_1 + n_2$. Here, X is a binary variable with value 1 if participant j is in the intervention group, and 0 otherwise, and $\sum_{j=1}^{n_1+n_2} X_j = n_2$. Be sure to use the likelihood given here, parameterized by (β_0, β^*) as the basis for this derivation. Begin by specifying H_0 in terms of β_0 and β^*.

   Note: One way of writing the "classic" χ^2 test is

$$Z^2 = \frac{(\hat{p}_1 - \hat{p}_2)^2}{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}},$$

   where $\hat{p}_1 = \frac{\sum_{j=1}^{n_1+n_2}(1-X_j)Y_j}{n_1}$ and $\hat{p}_2 = \frac{\sum_{j=1}^{n_1+n_2} X_j Y_j}{n_2}$.

The following two questions are based on MaxART data (Khan et al. 2020*).

* Khan, S., Spiegelman, D., Walsh, F., Mazibuko, S., Pasipamire, M., Chai, B., Reis, R., Mlambo, K.,

The data set "MaxART HW1 2026.RData" is in the Homework 1 folder in Canvas. The outcome variable is named "Y", where 1 means CD4 counts at 18 months were $\geq$ 350 cells/mm^3, and 0 otherwise, "X" is the treatment indicator, where 1 indicates that the clinic at enrollment was randomized to early access to ART as soon as testing positive to HIV, and 0 indicating the clinic at which the participant enrolled was randomized to standard of care.

3. Test H_0: the intervention has no effect on achieving CD4 counts $\geq$ 350 cells/mm^3 at 18 months vs. H_A: the intervention has an effect on achieving CD4 counts $\geq$ 350 cells/mm^3 at 18 months, by the Wald, score and likelihood ratio tests, using the logistic regression model. a)State the hypothesis mathematically. b) Perform the Wald, Score, and Likelihood ratio tests. c) Compare the results from the 3 test statistics in a concise, well-labeled table. d) Interpret your findings concisely from a subject matter point of view.

4. Calculate the a) power, b) sample size and c) MDE of the odds ratio (OR) in MaxART for, as relevant, power at 80% and 90%, and significance level at 5%, and use the estimated value of $\hat{\beta}^*_{H_A}$ observed in MaxART for these calculations. Use other background design features as found in MaxART as needed, and be sure to clearly specify all parameters assumed for each of the 3 calculations, using all 3 test statistics. Create a concise, well-labeled table summarizing results.

5. Required for BIS PhD students, optional for MS, MPH, PhD students in other departments, and other students: Derive all 3 asymptotic test statistics – a) the Wald, b) the score and c) the likelihood ratio tests – to compare binary outcomes in two groups

Delva, W., Khumalo, G., Zwane, M., Fleming, Y., Mafara, E., Hettema, A., Lejeune, C., Chao, A., Bärnighausen, T., & Okello, V. (2020). Early access to antiretroviral therapy versus standard of care among HIV-positive participants in Eswatini in the public health sector: the MaxART stepped-wedge randomized controlled trial. Journal of the International AIDS Society, 23(9), e25610.

based upon a Bernoulli likelihood with the log link function. The log link function means that the Bernouilli parameter, p, has form $p = E(Y|X_{ij}) = exp\{\beta_0 + \beta^* X_{ij}\}$.